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InsightAI dashboard

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Final Year Project

January 3, 2025

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Abstract

In today's data driven world, the power to quickly analyze data is crucial. Existing analysis tools are complex, costly and lack flexibility making them a bad choice for users. This creates a gap, hindering effective data-driven decision-making. InsightAI dashboard fills this gap by offering an application that helps users in performing analytics with ease at no cost, designed to empower users by simplifying complex analysis tasks. Intuitive platform let users gain insights from data without technical knowledge required. Application will support various file formats like CSV, SQLite databases and more. Famous libraries integrated to provide range of charts and analysis features. The AI-driven feature reduces time and effort required to analyze data allowing users to talk with data and gain information, helping them make informed decisions. Its power to streamline data visualizations, querying, and reporting makes it an everyday tool which improves user productivity and accessing full potential of their data.

Executive Summary

It is crucial for people such as business owners, researchers and data professionals that they can gain valuable and actionable insights from their data within not time and easily. As everything revolves around data, there is an increasing demand among people for applications that can help them in understanding and making use of their data. The products currently available in the market are costly and technical to use for many beginners in the field. These limitations hamper companies and individuals from accessing the true value of their data.

The InsightAI dashboard aims to solve this problem by filling the gap with an intuitive, easy to use and a powerful platform that solves the overly complex problem of data analysis. Our application was designed with accessibility in mind thus allowing users of every type of background to use our application with ease whether they be data professionals, researchers, or business owners. Our seamless data import, processing and visualization capabilities let users play with their data via natural language queries, generate sophisticated visualizations, and receive AI-driven insights. This all can be achieved by our application without the need for any sort of technical knowledge.

Our application supports various types of file formats such as CSV, Excel, and PDFs, and offers advanced data analytics features such as SQL query generation, customizable dashboards, and business intelligence tools. Furthermore, the integration of machine learning and natural language processing allows the users to easily extract useful information from their data making our application a cost effective and simple solution for business owners and individuals throughout the world that are looking for an application that can maximize the value of their data.

With an emphasis on usability, flexibility and timely insights our application aims to transform how people interact with their data by unlocking the true potential of their data without barriers and helping them make informed decisions, thus overall enhancing their productivity across all the sectors. With the creative features and the user-friendly design of the application, our application InsightAI stands as a game-changing solution for making data-driven decisions effortlessly.

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Chapter 1 Introduction

In today's world where data fuels progress, the power to quickly analyze and gain insights from data is of utmost importance to businesses and researchers. No one has the time to stare at vast amounts of data and try to generate useful insights from it. The options available right now in the market are paid, slow, complex for the new users and lack the flexibility option in them for the users. Our application InsightAI dashboard eliminates this problem by providing a powerful but still user-friendly and intuitive platform for data analysis that meets the needs of users with varying levels of expertise.

The application will feature a responsive and interactive interface for its users, making sure that the whole process could even be understood by a non-technical person. The backend of the application will be the area where the data processing and analysis will take place. It will feature popular libraries and frameworks for robust and quick performance.

Our application aims to automate and improve the way people carry out the data analytics tasks by using Artificial Intelligence (AI), Machine Learning (ML) and Natural Language Processing (NLP), eliminating the need for hiring expensive data analysts and engineers to extract useful insights from the data. By using our feature-rich platform, users will be able to talk with their data and get relevant information eliminating the need for expensive solutions. A key aspect of our application will be that users can select their desired columns for analysis, choose chart types and customize their visualizations without needing to have write lengthy codes. Our application aims to make the data analysis workflow smooth, from data import and cleaning to visualizing and reporting all within a single, integrated environment.

1.1 Purpose of this Document

The sole purpose of this document is to provide a comprehensive view of the application talking about the objectives, features and intended users for our application. This document serves as a reference for all users making sure alignment of the project goals, scope and terminology. It outlines the core components of the application from the user interface to the backend functionality while also defining the scope and limitations of the project. The end goal for this project is to provide clarity on how our application will help users in making informed, data-driven decisions, especially those users who do not possess technical expertise.

1.2 Intended Audience

The intended audience for this application encompasses a wide range of people who shall interact and use it in a variety of ways. These include:

- Business Stakeholders are interested in how our application can improve their informed data driven decisions in helping their organization grow.
- Non-technical users such as business owners, educators, researchers, or individuals will use our application as an easy-to-use tool for data analytics without the need for any type of technical knowledge.
- IT Professionals may use our application who require advanced functionalities, customizations, and SQL integration for more complex data analysis tasks.

By making sure that all people have the same level of access to our application we aid people in making insightful decisions using their data.

1.3 Definitions, Acronyms, and Abbreviations

To make sure that everything is clear, and also to avoid any type of confusion this section provides the key terms, acronyms and abbreviations used within this document:

Final Year Project (**FYP**)
Minimum Viable Product (**MVP**)
User Interface (**UI**)
User Experience (**UX**)
Agile Development (**Agile**)
Scrum Development (**SCRUM**)
Sustainable Development Goal (**SDG**)
Natural Language Processing (**NLP**)
Machine Learning (**ML**)
Structured Query Language (**SQL**)
Application Programming Interface (**API**)
Small and Medium Enterprises (**SMEs**)
Comma-Separated Values (**CSV**)

1.4 Conclusion

Our application is set to change the way businesses and individuals interact with data. By simplifying data analysis and integrating cutting edge AI technology in our application we empower users to gain information from their data regardless of their technical expertise. This document serves as the reference point for easy and accurate communication and an understanding of the application's purpose and scope.

Chapter 2 Project Vision

InsightAI Dashboard is a data analytics platform that is developed to provide data analysis accessible to people with different levels of expertise. It has the power to simplify complex analytical tasks by allowing the users access to upload their datasets and then process and gain valuable information from them there is also support for various multiple file formats that includes files such as CSV, Excel and PDFs. The application has the power to generate insights using natural language queries. With the application's built-in machine learning and customizable data visualizations users can quickly produce quality results on their datasets without having to require any technical skills. The application's intuitive and user-friendly interface combined with features such as report generation and diagram creation makes it an all-in-one tool for businesses, researchers, and individuals who are looking for free and user-friendly data analytics solutions.

2.1 Problem Domain Overview

InsightAI Dashboard revolutionizes the way people interact with their data. It provides a platform to the people for easy data analysis without having to require any technical background. The system is incorporated with a wide range of features which includes support for various type of file formats, customizable dashboards, statistical analysis and diagram generation. The main crux of this application is to reduce the time and effort that is required for data analysis hence making it accessible for both technical and non-technical users.

By the integration of business analytics and machine learning algorithms into our application we will not only be able to generate meaningful insights but will also be able to provide customized reports and visual representations that are easy to read and understand. This system serves two purposes providing easy solutions to complex data analysis tasks but will also improve business-related decision-making processes.

2.2 Problem Statement

In today's time there is a dire need for a data analysis tool that is user-friendly, cost-effective and accessible for all non-technical users. The tools that already are available in the market are expensive and complex to be used by many small businesses, entrepreneurs and casual users. They are unable to take full advantage of the data-driven decision-making. Our application aims to fill in this gap by providing a platform that simplifies data analysis making the non-technical users to be self-capable of performing complex analytics without technical expertise.

2.3 Problem Elaboration

The main problem lies in the complexity and cost of the current data analysis tools. Traditional tools such as Power BI, Tableau and others require a head start of learning before the user can start using them or significant financial investment. Also, many tools lack the user-friendly interface making them a very infeasible solution for many non-technical background users.

Sub-problems that arise from this include the complexity of usage as existing data analysis tools require specialized training to understand data querying, visualization, and analysis before someone starts to use them. They are often cost-prohibitive, requiring expensive licenses or subscriptions. Additionally, they have very poor interactivity since the user has to select a particular template or tool rather than the ability to manipulate a specific report or visualization with someone's requirements. Furthermore, they are also time-consuming as processing large datasets and then manually generating insights is an inefficient and error-prone task. Our project aims to eradicate these sub-problems to simplify the analysis process, provide intelligent insights, and allow users to communicate with their data intuitively.

2.4 Goals and Objectives

Our aim is to make an optimized and feature-rich application that helps users to effortlessly generate useful insights from their data. The following are the key goals and objectives for the development of our application:

- Develop a intuitive and user-friendly web application for comprehensive data analysis.
- Develop a powerful backend system integrated with AI that can manage data analytics and processing tasks.
- Include interactive data analysis and visualization tools.
- Enable users to develop various types of diagrams such as Flowcharts, Sequence, Class, State, ER diagrams and many more directly within the application.
- Provide users the ability to generate and execute SQL queries within the application.
- Multiple options of various types of data preprocessing.
- Business analytics tools to allow users to gain valuable business numbers and key insights from data.
- Customizable interfaces and themes for the user interface.
- User guidance to explain the features and working of our application to users.

- Optimization for handling data effectively.
- Introduce a feature to generate data summaries and produce insightful reports answering key questions about the data.
- Feature of uploading documents and inquiring questions from the application so that users can get to know about their document without the need for going through it.
- Uploading various file formats including PDF, DOC, TXT, RTF and many more.
- Advanced statistical insights will be displayed such as correlation matrix and more for quick business analysis.
- Contextual memory will be implemented for a more personalized experience and accurate responses to the user.
- Develop a customizable dashboard within the application for users to create, modify, and export visualizations and charts directly.
- Allow users to customize visualizations and seamlessly export them into the integrated dashboard, with options to download in their preferred format.

By integration of the above features, we aim to make our application a powerful solution for transforming data into actionable insights.

2.5 Project Scope

Our application will consist of the following components that will work harmoniously together to attain the project's desired goals.

- The front-end development will focus on creating a user-friendly, responsive, and intuitive interface that caters to both technical and non-technical users.
- The application will feature a dashboard where users can organize, manage, edit, and visualize their data, charts, and graphs easily.
- It will support uploading, processing, and storing various file formats, including CSV, Excel, SQLite databases, and document types such as PDF, DOC, and TXT.
- Natural language processing will be implemented for the chat interface, enabling context understanding and memory retention for personalized interactions with the application.
- Users will be able to add insights to the dashboard and manually edit their dashboards.
- The application will be powered by a suite of analysis tools, including SQL query generation,

statistical analysis, and business analytics features.

- The backend development will focus on optimizing data analysis, allowing for efficient handling of large datasets with an emphasis on quick response times and processing.
- Advanced algorithms and popular libraries will be used to generate the best insights and visualizations.
- The application will support the generation of various types of diagrams such as Flowcharts, Sequence, Class, State, ER diagrams, and more.
- There will be support for SQL querying, data pre-processing, and cleaning.
- The application will also provide document analysis, a customizable interface, and the ability to export user's work into different formats seamlessly.
- The application will include data summary and insight report generation, offering users key insights and relevant answers to their questions.

Our project will be developed using modern web development technologies such as NextJS for the front end and FlaskAPI or FASTAPI for the back end. We will be utilizing SQLite as our database and VectorDB for enhanced data retrieval. For styling and interactivity purposes, Tailwind CSS and TypeScript will be used. To ensure continuous feedback and iterative developments, Agile development practices will be used. While the project does not include server hosting or mobile application development at this phase, it will allow for future scalability and integration with external APIs.

2.6 Sustainable Development Goal (SDG)

InsightAI Dashboard aligns with several Sustainable Development Goals (SDGs) by offering a user-friendly and free to use platform that allows the users in making informed decisions based on the data, enhances the productivity of the users and promotes inclusivity in accessing advanced technology. Following are the relevant SDGs that the project directly supports:

2.6.1 SDG 8: Decent Work and Economic Growth

Our platform contributes to SDG 8 by empowering small business owners and individuals with a free to use and intuitive tool for analyzing and visualizing their data. Our platform enhances productivity be enabling the users to make data driven decisions without the need of extensive technical background or the need of hiring data scientists. By simplifying data analytics, our platform helps foster economic growth majorly for small business owners who do not possess the required capital to invest in expensive data analysis tools. Our platform promotes sustainable economic development by increasing operational efficiency, helping in making informed decisions and competitiveness.

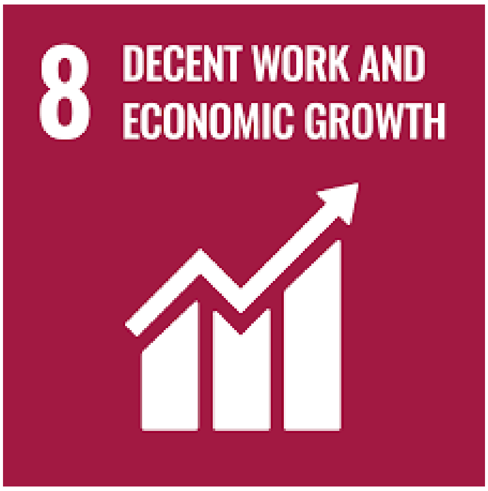

Figure 2.1: Decent Work and Economic Growth: InsightAI promotes innovation by offering easy-to-use data tools, supporting the development of modern industries

2.6.2 SDG 9: Industry, Innovation, and Infrastructure

Our platform aligns with SDG 9 by promoting innovation in data analytics and also modernizing how individuals and businesses interact with their data. Our platform integrates artificial intelligence (AI) and machine learning (ML) to allow the smooth flow of data analysis processes and providing users with a tool that helps them in extracting useful insights from their data effortlessly. By offering a cutting-edge solution that can be used across industries our application helps in the development of strong infrastructure and motivates the widespread adoption of innovative technologies in business and research sectors.

Figure 2.2: Industry, Innovation and Infrastructure: InsightAI helps small businesses make better decisions using data, leading to business growth and more job opportunities

The application plays a significant role in supporting multiple SDGs especially those which are oriented towards promoting economic growth and fostering innovation. Our free to use and intuitive platform for data analysis helps in empowering users of all backgrounds to use the power of their data for their decision-making processes thus contributing to sustainable development across industries.

2.7 Constraints

Our application while designed for robust data analysis and superior performance still does have certain constraints such as the following:

- It has a token limit and cannot process more than 15,000 tokens in a single interaction, which may hinder the handling of large datasets or very complex queries.
- The application can only process datasets up to a maximum of 500 MB.
- Users should be aware that on rare occasions it may produce errors in code or provide incorrect answers, thus approach the application with a close eye, especially in situations where accuracy is paramount.

While ambitious in scope certain advanced features, for example real-time data connections, integration with external APIs, and collaborative editing will be considered for future development phases.

2.8 Business Opportunity

Our application has a significant business opportunity in the data analytics space, as it is free to use, the data visualization and analysis has the power to disrupt the traditional, costly platforms available in the market. Small businesses, startups, researchers, and individuals who cannot purchase current expensive analytical tools will benefit greatly from our application.

2.9 Stakeholders Description/ User Characteristics

The success of our application lies in the fact that we understand the needs and characteristics of its users. In this section we will provide a detailed description of different types of users, how they will benefit from using our application, each with varying objectives, technical expertise and use cases. By identifying these users we will be better able to understand and address their problems by making sure that our application meets their demand and solves their problems. From small business owners to enterprise users our application aims to help a wide range of audience by providing them with an intuitive and efficient data analytics solution. Our application effectively addresses the business problems of stakeholders by providing them with insights through advanced features.

2.9.1 Stakeholders Summary

Small business owners are those users that need a cost-effective way to analyze their data to help them make quick and important business decisions. Researchers and academics are those users that need an intuitive tool to analyze the data and then generate insights from it. Casual users are those users that don't have the technical expertise required for data analysis but wish to understand and analyze personal data with ease. Enterprise users are large organizations that are looking for a cost-effective data analysis tool to complement or replace existing solutions in the market.

2.9.2 Key High-Level Goals and Problems of Stakeholders

The users of our application have respective goals and problems that they aim to address which includes small business owners who are in need for a low-cost and easy-to-use solution that can help them in making data-driven business decisions without hiring data scientists. Researchers require a flexible tool that gives them the room to analyze large datasets without the need for an extensive technical background. Casual users want to interact with their data intuitively, like using natural language processing to get to know about their data and generate visual reports quickly.

2.10 Conclusion

In conclusion, the vision of our project is making the process of data analysis and visualization efficient and userfriendly. We aim to revolutionize this industry with the use of AI. The project aims to let users carry out the data analysis and visualization tasks within a matter of seconds. Our project scope is to deliver a functional web application that includes all the features already addressed.

Chapter 3 Related Applications

In the rapidly growing landscape of data analytics understanding the existing solutions in the market and analyzing their limitations is crucial in developing an effective and innovative tool for our users. This chapter provides a comprehensive view of the relevant solutions that are available in the market with a focus on their functionalities, strengths and weaknesses. This chapter also states the definitions, acronyms and abbreviations used throughout the chapter for better understanding. By critically assessing the solutions we aim to gather functionalities for our users and also emphasize the need for an advanced analytics solutions that can better serve the evolving needs of businesses in today's data-driven environment.

3.1 Definitions, Acronyms, and Abbreviations

The following are the important definitions, acronyms and abbreviations used throughout in this section:

- Artificial Intelligence (**AI**)
- Business Intelligence (**BI**)
- Key Performance Indicator (**KPI**)
- Predictive Analytics (**PA**)
- Data Visualization (**DV**)
- Application Programming Interface (**API**)
- Structured Query Language (**SQL**)
- Machine Learning (**ML**)
- Software as a Service (**SaaS**)
- User Experience (**UX**)
- Decision Support System (**DSS**)

3.2 Detailed Literature Review

This section includes an overview of the related applications in the data analytics space. Focusing on their functionalities, strengths and weaknesses. By examining this we aim to come up with a application that is better suited and meets the requests of our users.

3.2.1 Databox Application

3.2.1.1 Summary

Databox [1] is a KPI dashboard platform that is designed for businesses that aim to collect and display key performance indicators (KPIs). It integrates numerous platforms to gather metrics like sales, leads and conversions all in one central dashboard. It updates data automatically. Users can also create custom dashboards and choose from various visualizations.

3.2.1.2 Critical Analysis

Databox excels in providing real time access to key performance indicators (KPIs) and seamless integration with multiple data sources. It simplifies data retrieval and makes intuitive dashboards. However, Databox has notable limitations when it comes to advanced analytics. It lacks AI-driven features for deeper analysis. There is no machine learning which restricts its usefulness in complex business scenarios where insights from large datasets need to be uncovered beyond basic metrics.

3.2.1.3 Relationship to the Proposed Application

Compared to InsightAI the Databox application does not provide advanced AI-based analytics. Our application also incorporates machine learning algorithms to generate business insights, uncover hidden patterns and perform complex data analysis. InsightAI has advanced AI capabilities that fill this gap by automating data analysis, provide predictions, and generate reports that can be directly used for decision making. InsightAI stands as a more comprehensive solution for data analysis than Databox.

3.2.2 Geckoboard Application

3.2.2.1 Summary

Geckoboard [2] is a dashboard software that empowers businesses to visualize their data easily and effectively. It helps in data-driven decision making. Geckoboard connects to over 90 data sources to create live dashboards ensuring that key metrics are visible and understandable at a glance.

3.2.2.2 Critical Analysis

Geckoboard addresses common data accessibility issues by eliminating the need for complex BI tools or data specialists. Its user-friendly interface allows create live dashboards, but it lacks AI capabilities. It falls short for organizations seeking deeper insights or predictive analytics.

3.2.2.3 Relationship to the Proposed Application

Unlike Geckoboard, which does not utilize AI and requires manual data handling and filtering our application uses advanced AI-driven analytics to automate data processing, generate insights and enhance visualization. This makes our applciation more suitable for businesses needing comprehensive and automated data analysis.Geckoboard is focused on straightforward data visibility.

3.2.3 Visual.IS Application

3.2.3.1 Summary

Visual.IS [3] is an online dashboard creator that is made to help businesses organize and then visualize their data efficiently. Users can also manually input data or import it from spreadsheets. The platform offers various widgets, chart types and the ability to filter and customize dashboards with themes. Features include multi-page dashboards, one-click sharing, embedded code options and tools for data management.

3.2.3.2 Critical Analysis

Visual.IS provides the users with customizable visualizations and user-friendly features but it lacks the AI capabilities for advanced analytics and insight generation. Additionally, it also has limitations such as only a maximum of 10,000 rows and 30 columns of data can be input into the application which restricts its use for larger datasets.

3.2.3.3 Relationship to the Proposed Application

In contrast to Visual.IS our application the InsightAI dashboard utilizes advanced AI-driven analytics to automate data processing, generate insights and visualize data with ease. This capability allows businesses to analyze larger datasets without requiring any manual power making our application a more suitable choice for organizations.

3.2.4 Dashboard Builder Application

3.2.4.1 Summary

Dashboard Builder [4] is a tool for creating dynamic data-driven dashboards that can be installed either on-premises or accessed online. It allows visualization of data through various chart types. The platform supports drag-and-drop functionality and integrates with multiple data sources, providing a SQL query builder for advanced data manipulation. Users can build responsive dashboards and generate HTML or

PHP code for embedding but interactivity in charts is limited which hinders user engagement with the data.

3.2.4.2 Critical Analysis

While Dashboard Builder offers a robust set of features for visualizing data its high cost—starting at 196 dollars per month for the kickstarter plan makes it less accessible for small businesses or teams with limited budgets. The absence of AI features and non-interactive charts restricts its capability to provide dynamic data exploration. Users cannot share dashboards easily which limits collaborative decision making.

3.2.4.3 Relationship to the Proposed Application

In contrast to Dashboard Builder, InsightAI uses AI-driven analytics to automate data processing and generate actionable insights facilitating a more efficient decision-making process.

3.2.5 Infogram Application

3.2.5.1 Summary

Infogram [5] is a platform to create interactive data visualizations. It has a vast library of over 35 chart types and 800 map styles with the ability to import data from various sources. Infogram offers real-time data updates, enabling users to keep their visualizations current.

3.2.5.2 Critical Analysis

While Infogram has a user-friendly interface and extensive design options, it is complex in layout making it challenging for some users to navigate effectively. The platform requires significant amount of manual input and customization, especially for detailed projects. This complexity can detract the user experience particularly for those seeking straightforward solutions for data visualization.

3.2.5.3 Relationship to the Proposed Application

In comparison to InsightAI, which uses advanced AI to automate data insights and reporting Infogram's dependency on a complex layout and manual customization is a drawback. Our application provides seamless data integration without the need for extensive manual handling. Additionally, our application focuses on delivering actionable insights efficiently, enhancing decision-making processes.

3.2.6 Klipfolio Application

3.2.6.1 Summary

Klipfolio [6] is a data analytics and dashboard platform designed to help teams track metrics, analyze data and create customizable dashboards and reports. Klipfolio integrates with hundreds of service APIs and data sources, allowing teams to build real-time dashboards and manage live data. The platform also offers two main products: PowerMetrics which focuses on putting data analysis into the hands of business users and Klips which are highly customizable dashboards and reports.

3.2.6.2 Critical Analysis

Klipfolio serves as a robust platform for visualizing data and tracking metrics but its absence of AI features limits its ability to deliver AI-driven insights. The platform's dependency on manual data integration and lack of predictive analytics hinders businesses looking for tools to automate insights and streamline decision-making. Klipfolio's pricing structure is also a disadvantage for smaller businesses and teams especially considering its lack of AI-based automation.

3.2.6.3 Relationship to the Proposed Application

In comparison to InsightAI, which offers advanced AI-driven analytics Klipfolio falls short in terms of automation and intelligent data insights. Our application enables users to generate insights autonomously and create reports without manual intervention whereas Klipfolio lacks these AI capabilities. Furthermore, our application focuses on automated decision-making which makes it a more powerful tool for research work that requires quick and accurate data analysis.

3.2.7 Zoho Analytics Application

3.2.7.1 Summary

Zoho Analytics [7] is an AI-powered self-service BI and analytics platform that allows businesses to go from data to insights. It gives comprehensive tools for data connection, preparation, analysis and visualization. It takes data from over 500 sources. The platform supports predictive analytics, anomaly detection and automation features making it suitable for a variety of business use cases. Zoho Analytics provides interactive, visually engaging dashboards with 50+ visualization options, offering collaborative tools for teams, and embedded BI features for seamless integration with business ecosystems.

3.2.7.2 Critical Analysis

While Zoho Analytics offers many features its AI capabilities remain secondary to its core functionality of reporting and visualization. The platform is accessible to both technical and non-technical users but still its complex interface presents a learning curve especially for those unfamiliar with BI tools. While the pricing begins at a reasonable 25 dollars per month it quickly escalates based on the scale of usage, which poses challenges for smaller businesses and startups. Zoho struggles with the management of large datasets, limiting its application for enterprises requiring large-scale data analysis.

3.2.7.3 Relationship to the Proposed Application

When compared to InsightAI, which focuses on advanced AI-driven analytics, automated insights and predictive capabilities Zoho Analytics fails in terms of AI sophistication. Our application uses deep AI integration to automate data insights making it more suitable for complex research and business intelligence tasks that require minimal manual intervention. The limited AI in Zoho restricts its usage to basic reporting tasks whereas our application excels in offering a more hands-off approach to data analysis.

3.2.8 ClicData Application

3.2.8.1 Summary

ClicData [8] is a data integration and visualization platform that offers tools for data management, analytics and reporting. It allows users to connect over 200 cloud and on-premise applications and databases, cleanse and transform data through built-in templates and visualize data through dynamic dashboards. ClicData also provides features for report automation, real-time alerts and sharing through various formats.

3.2.8.2 Critical Analysis

The platform lacks advanced AI capabilities. In today's data-driven environment AI is increasingly crucial for extracting deeper insights and automating complex analytics processes. ClicData's absence of AI limits users' ability to perform predictive analysis, detect patterns, or automatically interpret large volumes of data in real-time. The platform is also geared more towards standard reporting making it effective for static analysis but less suitable for organizations that require dynamic, real-time insights or advanced data analytics powered by AI. Moreover, while the basic plan is accessible at 39 dollars per month the platform's cost may escalate with additional features and users handling larger datasets may encounter performance bottlenecks.

3.2.8.3 Relationship to the Proposed Application

Compared to InsightAI, ClicData offers a much more basic framework for data analysis, focusing on reporting and dashboard creation without any AI integration. Our application excels in AI-powered analytics offering features such as predictive insights and automated reporting.

3.2.9 DashThis Application

3.2.9.1 Summary

DashThis [9] is an automated reporting tool designed specifically for marketers offering an intuitive platform to create visually appealing reports effortlessly. It connects to major digital marketing platforms allowing users to compile their key performance indicators (KPIs) into comprehensive dashboards. The service emphasizes time-saving features, such as automated report generation and customizable templates enabling marketers to streamline their reporting processes significantly. DashThis aims to facilitate efficient data analysis and reporting for a wide array of users, including marketing teams, agencies, and small businesses.

3.2.9.2 Critical Analysis

It lacks AI functionalities that could enhance data insights and analysis limiting its effectiveness in predictive analytics. While the basic plan starts at 39 dollars per month which might be reasonable for established businesses, it proves expensive for startups and small enterprises. Moreover, the customization options are somewhat limited and users find the absence of interactive charts a drawback when presenting data dynamically.

3.2.9.3 Relationship to the Proposed Application

When compared to InsightAI, which excels in AI-driven analytics and advanced data insights DashThis presents a more basic reporting solution. Our application's focus on automation distinguishes it from DashThis particularly for businesses that require comprehensive, AI-enhanced insights.

3.2.10 Tableau Public Application

3.2.10.1 Summary

Tableau Public [10] is a free platform for individuals to explore, create and share data visualizations online. It encourages users to showcase their analytical skills and engage with the data visualization community through features like “Viz of the Day” and an extensive portfolio system. Users can access a diverse array of data visualizations. Tableau Public is designed to facilitate creativity and collaboration among data enthusiasts while providing tools for effective data storytelling.

3.2.10.2 Critical Analysis

While Tableau Public is widely recognized for its user-friendly interface and extensive library of shared visualizations, it presents several limitations. It lacks integrated AI features which restricts users from performing advanced data analysis or predictive modeling. The platform does offer a free version, making it accessible for users looking to familiarize themselves with data visualization however, the costs associated with its full version can be quite high for organizations seeking advanced features. Furthermore, one significant drawback is that all data published on Tableau Public is publicly accessible which may not meet with the needs of businesses that require confidentiality and secure data management.

3.2.10.3 Relationship to the Proposed Application

When comparing Tableau Public to InsightAI, distinct differences emerge particularly regarding their core functionalities and focus. Our application offers advanced AI-driven analytics that enables users to perform complex data analysis, generate actionable insights and automate reporting processes distinguishing it from Tableau Public's visualization centric approach. Our application focuses on AI integration providing a more comprehensive solution for organizations seeking deeper analytical capabilities and predictive modeling. Although Tableau Public is free to use, its lack of AI features and the public nature of shared data can hinder its effectiveness for businesses requiring confidentiality and advanced insights.

3.3 Literature Review Summary Table

3.4 Conclusion

This chapter analyzed various data analytics applications, highlighting their functionalities, strengths and limitations. While tools like Databox, Geckoboard, and Tableau Public excel in data visualization and reporting, they often lack advanced AI-driven capabilities essential for deeper insights and automated analytics. In contrast, InsightAI addresses these gaps by offering sophisticated AI algorithms which facilitates automated data processing and actionable insights making it a better choice for businesses looking to navigate the complexities of today's data-driven environment.

Table 3.1: This summary table analyzes various data analytics applications, detailing their key features and limitations while demonstrating how our application addresses these shortcomings

Application	Key Features	Limitations	How Our Application is Better
Databox [1]	KPI dashboard platform, integrates with multiple data sources, real-time updates	Lacks AI-driven analytics, no machine learning, limited metrics, no manual data handling	Includes AI insights, automated cleaning, supports unstructured files, larger uploads.
Geckoboard [2]	Live dashboards, connects to 90+ data sources, user-friendly	No AI capabilities, lacks predictive analytics, manual data handling	Adds AI features, predictive analytics, research assistant, handles unstructured data.
Visual.IS [3]	Customizable dashboards, various widgets, multi-page dashboards	No AI capabilities, limited to 10,000 rows and 30 columns, manual data handling	Supports AI analytics, relational diagrams, automated cleaning, handles larger datasets.
Dashboard Builder [4]	Dynamic dashboards, SQL query builder	Expensive, lacks AI and interactivity, limited data handling	Offers AI insights, automated data handling, more affordable for small businesses.
Infogram [5]	Interactive visualizations, real-time updates	Complex layout, significant manual input, limited data handling	Automated customization options, simpler interface, AI integration.
Klipfolio [6]	Real-time dashboards, customizable reports	No AI capabilities, manual integration, expensive for small businesses	Offers AI insights, automated integration, handles unstructured files.
Zoho Analytics [7]	AI-powered BI, predictive analytics	Complex interface, limited large dataset handling, high pricing	Simplified interface, efficient handling of large datasets, lower prices.
ClicData [8]	Data management, analytics, reporting	Lacks AI capabilities, performance issues with large datasets	Incorporates AI, handles large datasets efficiently, more cost-effective.
DashThis [9]	Automated reporting for marketers	Lacks AI functionalities, limited customization, expensive	AI-driven insights, greater customization, free plans for small businesses.
Tableau Public [10]	Free platform for data visualization	No AI features, data is public, high costs for full version	Offers AI features, supports private data, more affordable for advanced features.

Chapter 4 Software Requirement Specifications

This chapter presents the software requirements for the development of InsightAI dashboard. It will cover the essential features such as functional and non-functional requirements, quality attributes, use cases, hardware and software requirements and lastly risk analysis. By going through this chapter, developers and users will gain a understanding of the system behavior, architecture and specifications needed to build and maintain it.

4.1 List of Features

InsightAI dashboard aims to revolutionize the way people carry out their data analysis and visualization tasks by providing the following features along with basic sign up/login functionality as a web application:

- **User-Friendly Web Application**: Our web application will offer a visually appealing and easy-to-use interface for non-technical and technical users alike.
- **Interactive Data Analysis**: Our application will provide interactive tools for analysis and data manipulation.
- **Diagram Creation**: Users can create flowcharts, sequence diagrams, class diagrams, state diagrams, ER diagrams, and more.
- **SQL Query Support**: Users can write and execute SQL queries directly within the platform.
- **Data Preprocessing**: Our application offers multiple data preprocessing techniques to users.
- **Customizable UI**: Offers customizable user interfaces and themes for personalization.
- **Data Ingestion**: Users can upload data into the web application.
- **AI-Powered Insights**: Users can generate automated insights and trends based on data analysis.
- **Visualization Tools**: Our application will display results using charts, graphs, and other visual aids for ease of understanding.
- **Custom Reports**: Users can generate and export custom reports.
- **Natural Language Processing (NLP)**: Our application allows users to interact with their data by asking questions with our AI-assistant without requiring any sort of technical knowledge.

4.2 Functional Requirements

This section highlights the functional requirements of our system which relate to its technical functionalities. These requirements include:

- Users can create accounts and log in securely to the application.
- Users can upload various file formats such as CSV, Excel, JSON, PDF, DOC etc to perform analysis.
- Users can input and execute SQL queries within the application to manage and manipulate datasets.
- Users can develop various diagrams such as Flowcharts, Sequence Diagrams, Class Diagrams, ER diagrams etc within the application.
- Users should be able to visualize data in various formats including graphs, pie charts, bar charts and more.
- Users can access tools that provide insights from data such as statistical summaries and trends.
- The system remembers user preferences and provides personalized suggestions based on previous interactions.
- Users can create and modify dashboards. They can also export visualizations and download them.
- Users can upload documents and analyze their data using AI by generating insights without manually reading the files.
- Users should be allowed to apply dynamic filters to datasets and observe data for insights.

4.3 Quality Attributes

InsightAI dashboard aims to become a secure, reliable and user-friendly application. Quality attributes are integral to the performance evaluation of a software system. They provide a checklist that aids in assessing the quality of the software. The quality attributes of our system are:

4.3.1 Usability

The interface of the application should be inviting and the users, in particular, should not have to spend time learning how to work on the application. It should have a friendly and intuitive design with easily understandable instructions. Its interface should be rather clear and imply the options for both ordinary users and for experts. The objective is to obtain the condition in which users can easily and efficiently solve their problems without much disorientation.

4.3.2 Scalability and Reliability

The idea is that the application should remain easy to change and improve through modular elements that are easy to maintain and update, so that the application can grow to address new demands from users or new additions to the set of features that the application should offer. It should be able to grow the system without necessarily impacting on the efficiency of the system. Also, it must be reliable which means that the application has to be always on-line and almost never fail, or in case of failure, quickly and effectively fix the problem.

4.3.3 Performance and Security

There are expected and unforeseen errors. Hence, the application should perform its functionalities efficiently with minimal errors, from data input, processing, analysis and report generation. It has to be versatile to separately process large quantities of data or numerous complicated operations resulting in efficient usage for the client. Also in this regard, the application must feature strong security features especially the user login that will ensure secure storage of the data collected by the application, ensure the privacy of the users at all times and build trust with the users all through its use. These security aspects should not in any way, affect the performance or usability of the system.

4.4 Non-Functional Requirements

This section will be discussing non-functional requirements for the different entities of InsightAI dashboard.

- The application will run reliably without unexpected failures to ensure users can access the application at all times.
- The application will maintain data integrity by preventing data loss or corruption during transactions and interactions
- The application will have proper access controls and authorization mechanisms, ensuring that users can only access their own data.
- The application will have an intuitive and user-friendly interface for users to perform tasks efficiently.
- The application will provide instructions guiding users through the application.
- The application must ensure maximum possible up-time, with the goal of achieving at least 99
- The web-based application must run across various devices and browsers.

4.5 Assumptions

During the design phase, some important assumptions were taken. Here are a few of them:

- Users are expected to possess a reliable and consistent internet connection.
- Users are presumed to have access to a device capable of running web applications.
- Users are assumed to possess a basic understanding of website navigation and usage.

4.6 Use Cases

This section shows examples of how some features of our system can be used.

4.6.1 User Sign Up

<table>
<tr><td>Name</td><td colspan="4">User Sign Up</td></tr>
<tr><td>Actors</td><td colspan="4">New User</td></tr>
<tr><td>Summary</td><td colspan="4">The new user creates an account on our application.</td></tr>
<tr><td>Pre-Conditions</td><td colspan="4">The user must already have a email account to register in application database.</td></tr>
<tr><td>Post-Conditions</td><td colspan="4">The user is successfully registered and would be redirected to the sign in page.</td></tr>
<tr><td>Special Requirements</td><td colspan="4">None</td></tr>
<tr><td colspan="5">Basic Flow</td></tr>
<tr><td colspan="2">Actor Action</td><td colspan="3">System Response</td></tr>
<tr><td>1</td><td>The user opens the sign up page.</td><td>2</td><td colspan="2">The sign up page is displayed asking for email and password.</td></tr>
<tr><td>3</td><td>The user enters valid email and password.</td><td>4</td><td colspan="2">The system verifies the email and password and then creates a account for the user and redirects the user to sign in page.</td></tr>
<tr><td colspan="5">Alternative Flow</td></tr>
<tr><td>5</td><td>The user enters invalid email or password.</td><td>6</td><td colspan="2">The system responds with an error message: Incorrect email or password entered.</td></tr>
</table>

4.6.2 User Sign In

Name	User Sign In		
Actors	User		
Summary	The user shall provide their email and password on the sign in form and after successful verification, redirect the user to the home page.		
Pre-Conditions	The user must be in the database records that is he should have already signed up with a account also the user must not already be logged in.		
Post-Conditions	The user's session is successfully established and shall be redirected to the home page.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user opens the sign in page.	2	The sign in page is displayed asking for email and password.
3	The user enters valid email and password.	4	The application verifies the email and password then establishes a session for the user and redirects the user to the home page.
Alternative Flow			
5	The user enters invalid email or password.	6	The application responds with an error message: Incorrect email or password entered.

4.6.3 User Sign Out

Name	User Sign Out		
Actors	User		
Summary	The user presses the sign-out button, which redirects the user to the sign-in page of the app. The user's profile is signed out simultaneously by the app.		
Pre-Conditions	The user must be in the database records that is he should have already signed up with a account and is logged in.		
Post-Conditions	The user's session is terminated, and the user is redirected to the sign-in page.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user presses the sign-out button.	2	The user is logged out by the app, and the sign in page is displayed asking for email and password.
Alternative Flow			
3	None	4	None

4.6.4 Upload New Files

Name	Upload New Files		
Actors	User		
Summary	User clicks on the file upload button from where a menu appears then the user selects the file and clicks on the upload button. The file is uploaded onto the application for use.		
Pre-Conditions	The user must be signed into the application.		
Post-Conditions	The file is uploaded onto the application and is shown to the user.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user presses the file upload button.	2	The file upload box is opened
3	The user selects the desired file and presses the upload button.	4	The file is uploaded into the app
Alternative Flow			
5	The user uploads a file that is in an unsupported format.	6	The application responds with an error message: Incorrect file format uploaded.

4.6.5 Delete Files

Name	Delete Files		
Actors	User		
Summary	User clicks on the delete files button from where the current uploaded files are shown then the user can select the files he want to delete and then click on the Delete Files button to remove them from the application.		
Pre-Conditions	The user must be signed into the application and files should be uploaded into the application.		
Post-Conditions	The file is deleted from the application.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user presses the Delete Files button.	2	The file selection box is opened
3	The user selects the desired files and presses the delete button.	4	The file is removed from the app
Alternative Flow			
5	None	6	None

4.6.6 Analyze Data

Name	Analyze Data		
Actors	User		
Summary	User clicks on the analyze data button then a series of different chart options are presented and the real-time view of the chart is also shown to the user. The user can also then export these charts to his dashboard by clicking on the export button.		
Pre-Conditions	The user must be signed into the application and files should be uploaded into the application.		
Post-Conditions	Different types of charts are shown to the user.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user presses the analyze data button.	2	Different types of charts are shown to the user
3	The user selects the desired chart and presses the export button.	4	The chart is moved to the dashboard.
Alternative Flow			
5	None	6	None

4.6.7 Data Visualization

Name	Data Visualization		
Actors	User		
Summary	User selects data visualization from the AI chat section of the application, then the user can ask the AI to generate different types of charts from the data and gain useful insights from them.		
Pre-Conditions	The user must be signed into the application and files should be uploaded into the application.		
Post-Conditions	The application generates and presents the user with the requested chart or visualization.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user selects the data visualization option and asks the application to generate any type of visualization or chart.	2	User is presented with the requested visualization.
Alternative Flow			
3	The user requests for a visualization but due to Internet issue the contact is broken.	4	The user is presented with an error message.

4.6.8 Research Assistant

Name	Research Assistant		
Actors	User		
Summary	User selects research assistant option from the AI chat section of the application, then the user can ask the AI to tell about the PDF document uploaded and many more things also it can ask the application to generate a report on the PDF.		
Pre-Conditions	The user must be signed into the application and files should be uploaded into the application.		
Post-Conditions	The application generates and presents the user with the requested data or report.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user selects ”Research Assistant” to inquire about the PDF or generate a report.	2	User is presented with the requested information or report.
Alternative Flow			
3	The user’s request for a report or information fails due to an internet issue.	4	The user is presented with an error message.

4.6.9 SQL Query

Name	SQL Query		
Actors	User		
Summary	User selects SQL query option from the AI chat section of the application, then the user can ask the AI, SQL queries, the application will give the answer to the SQL query results.		
Pre-Conditions	The user must be signed into the application and files should be uploaded into the application.		
Post-Conditions	The application generates and presents the user with the requested SQL queries results.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user selects ”SQL Query” to request query results from the application.	2	User is presented with the requested results of query.
Alternative Flow			
3	The user’s query request fails due to an internet issue.	4	The user is presented with an error message.

4.6.10 Data Cleaning

Name	Data Cleaning		
Actors	User		
Summary	User selects data cleaning option and asks the application to preprocess the dataset in multiple ways, the application will preprocess the dataset in the required manner and present the user with the results.		
Pre-Conditions	The user must be signed into the application and files should be uploaded into the application.		
Post-Conditions	The application preprocesses and presents the user with the cleaned dataset.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user selects ”Data Cleaning” to preprocess the dataset.	2	User is presented with the preprocessed dataset.
Alternative Flow			
3	The user requests for preprocessing of dataset but due to Internet issue the contact is broken.	4	The user is presented with an error message.

4.6.11 Business Analytics

Name	Business Analytics		
Actors	User		
Summary	User selects business analytics option from the chat section of the application, then the user asks the application to generate different business insights from the data and gain useful information from them.		
Pre-Conditions	The user must be signed into the application and files should be uploaded into the application.		
Post-Conditions	The application generates and presents the user with different business insights.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user selects the business analytics option and asks the application to generate business insights.	2	User is presented with the requested insights.
Alternative Flow			
3	The user requests for business insights but due to Internet issue the contact is broken.	4	The user is presented with an error message.

4.6.12 Mermaid Diagram

<table>
<tr><td>Name</td><td colspan="3">Mermaid Diagram</td></tr>
<tr><td>Actors</td><td colspan="3">User</td></tr>
<tr><td>Summary</td><td colspan="3">User selects mermaid diagram option from the chat section of the application, then the user can ask the application to generate different types of diagrams such as class diagram, sequence diagram and many more from the data.</td></tr>
<tr><td>Pre-Conditions</td><td colspan="3">The user must be signed into the application and files should be uploaded into the application.</td></tr>
<tr><td>Post-Conditions</td><td colspan="3">The application generates and presents the user with the diagrams.</td></tr>
<tr><td>Special Requirements</td><td colspan="3">None</td></tr>
<tr><td colspan="4" align="center">Basic Flow</td></tr>
<tr><td colspan="2" align="center">Actor Action</td><td colspan="2" align="center">System Response</td></tr>
<tr><td>1</td><td>The user selects the mermaid diagram option and asks the application to generate diagrams from the data.</td><td>2</td><td>User is presented with the requested diagrams.</td></tr>
<tr><td colspan="4" align="center">Alternative Flow</td></tr>
<tr><td>3</td><td>The user requests for a diagram but due to Internet issue the contact is broken.</td><td>4</td><td>The user is presented with an error message.</td></tr>
</table>

4.7 Hardware and Software Requirements

This section will highlight the hardware and software requirements which will make sure that the application efficiently provides ease to its users and makes sure that their data analysis and visualization journey a memorable experience.

4.7.1 Hardware Requirements

InsightAI dashboard hardware requirements have been limited to a minimum to allow this application to be accessed across multiple devices. The application possesses the power to be run on devices with moderate processing power and memory. The application is designed to be light weight and require users to have a stable internet connection to use the application uninterruptedly.

4.7.2 Software Requirements

The application requires that users run it on modern and up-to-date web browsers for a flawless experience and enhanced security. Users should enable cookies and cache functionalities in their browsers to improve the performance and user experience of the application.

4.8 Graphical User Interface

The GUI mockups for each screen along with details are provided in this section.

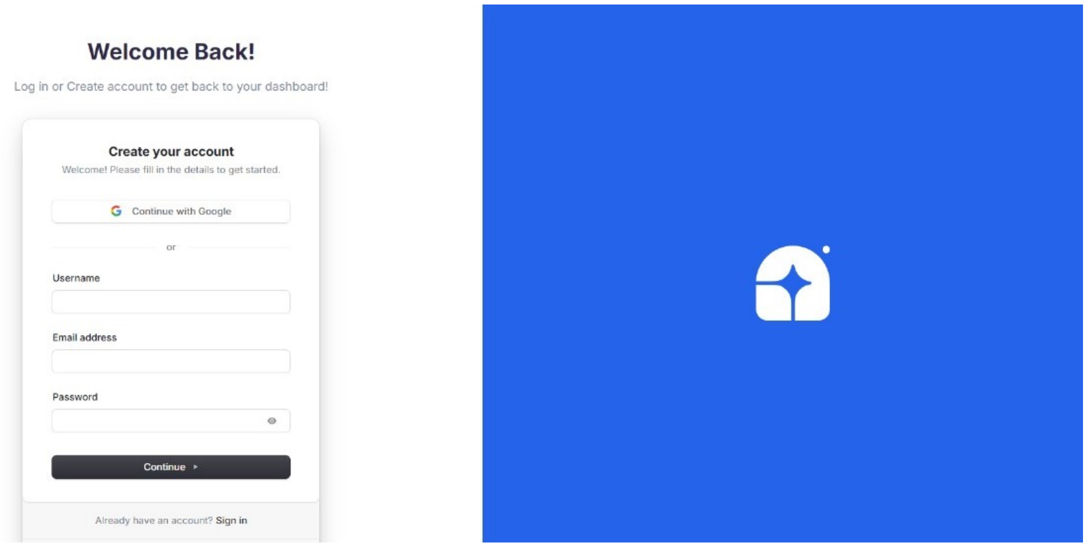

Figure 4.1: Sign-Up Page: This figure represents the user registration interface of InsightAI where new users will sign-up and create a new account

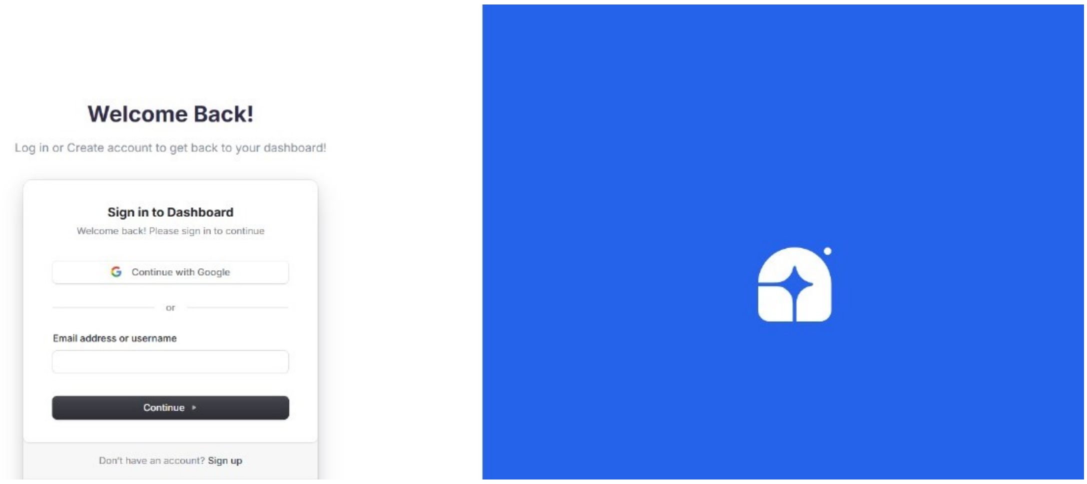

Figure 4.2: Sign-In Page: This figure represents the login interface of InsightAI from where existing users will login into their accounts

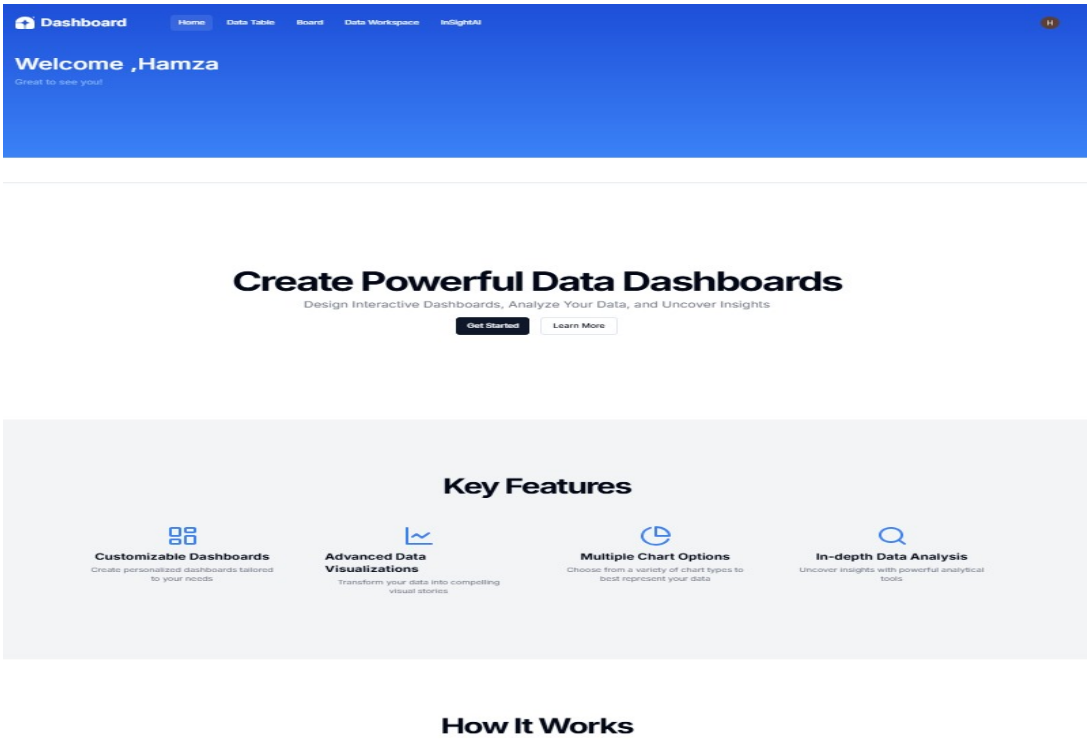

Figure 4.3: Home Page 1: This figure represents the main dashboard overview of InsightAI highlighting the application features

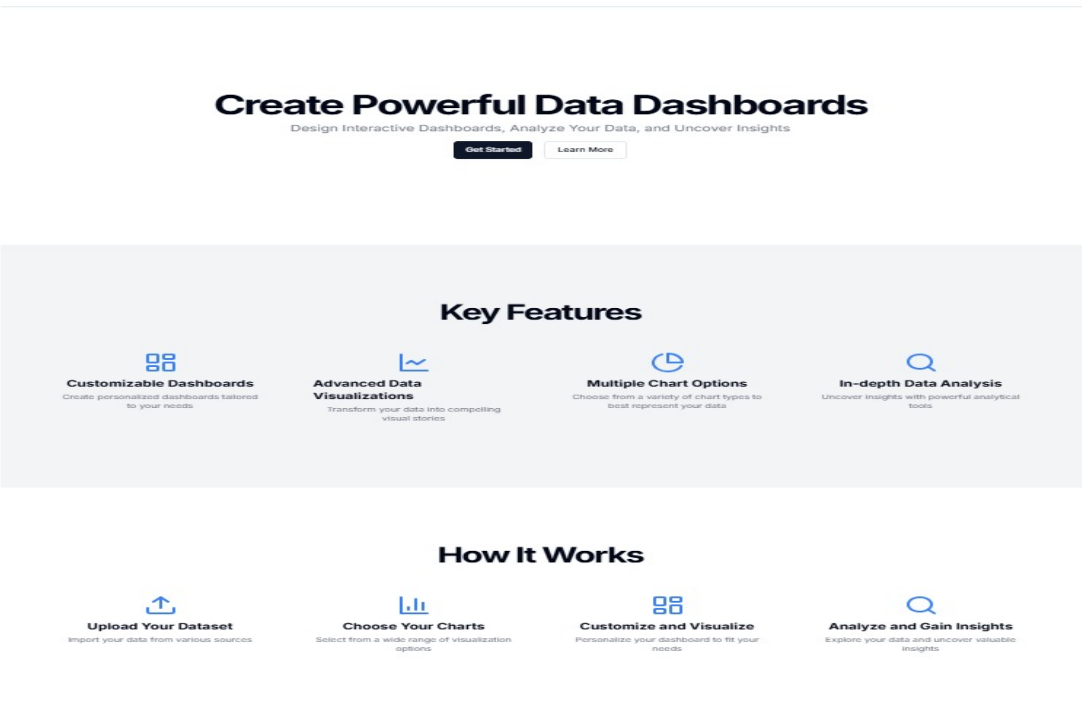

Figure 4.4: Home Page 2: This figure represents a secondary overview of the main dashboard page of InsightAI with some of the additional features

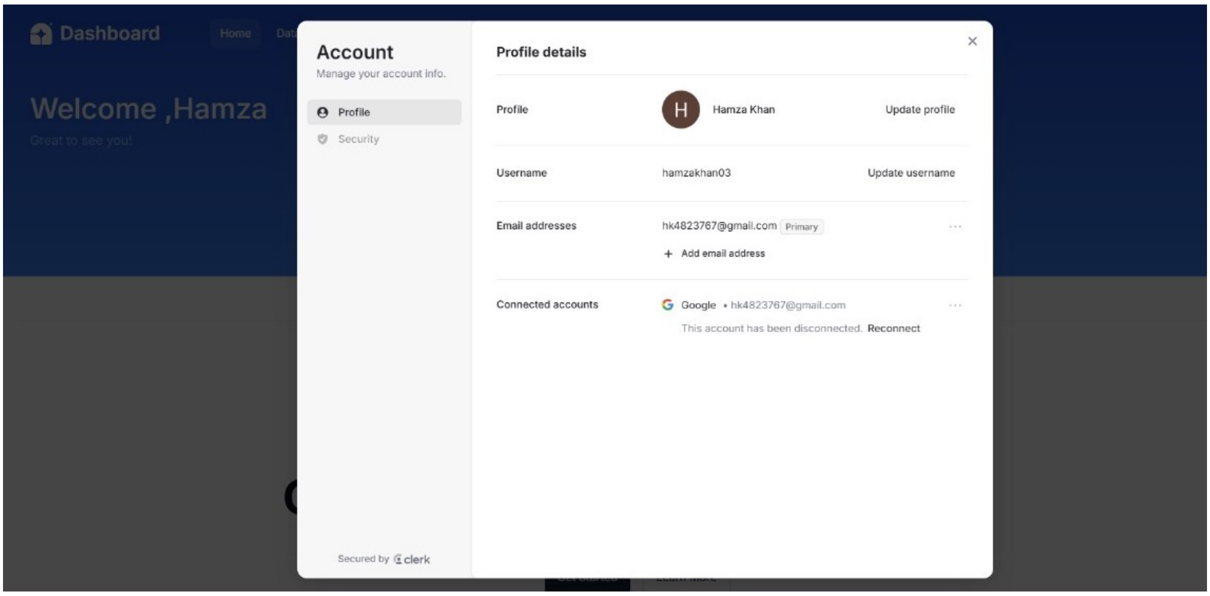

Figure 4.5: Account Settings Page: This figure represents the interface for managing different user account details of the InsightAI application

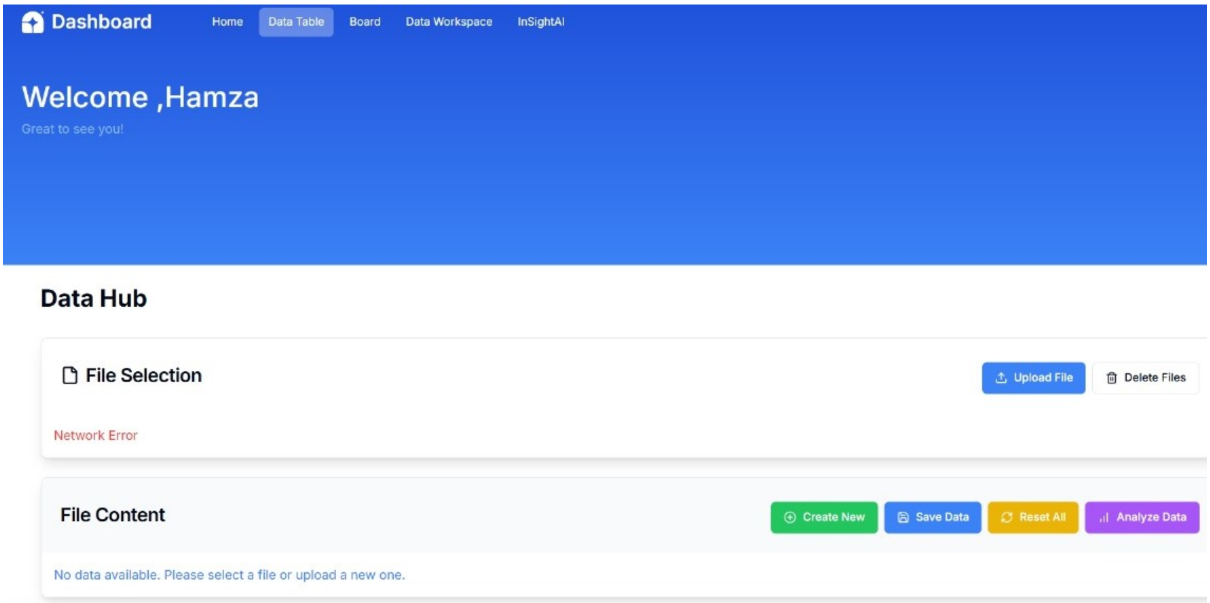

Figure 4.6: File Upload Page: This figure represents the interface for uploading files and data onto the InsightAI application

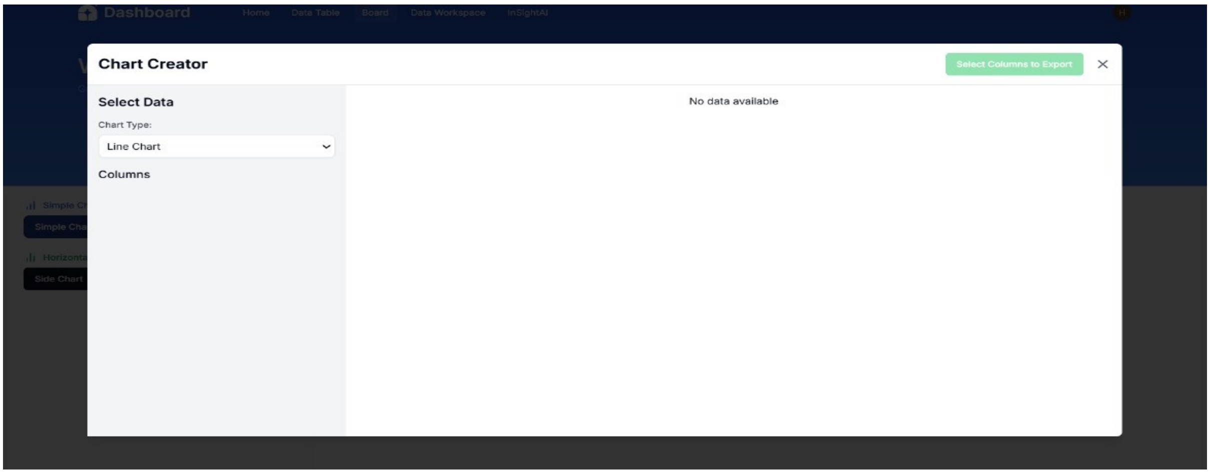

Figure 4.7: Chart Creator Page: This figure represents the interface for creating and customizing charts within InsightAI

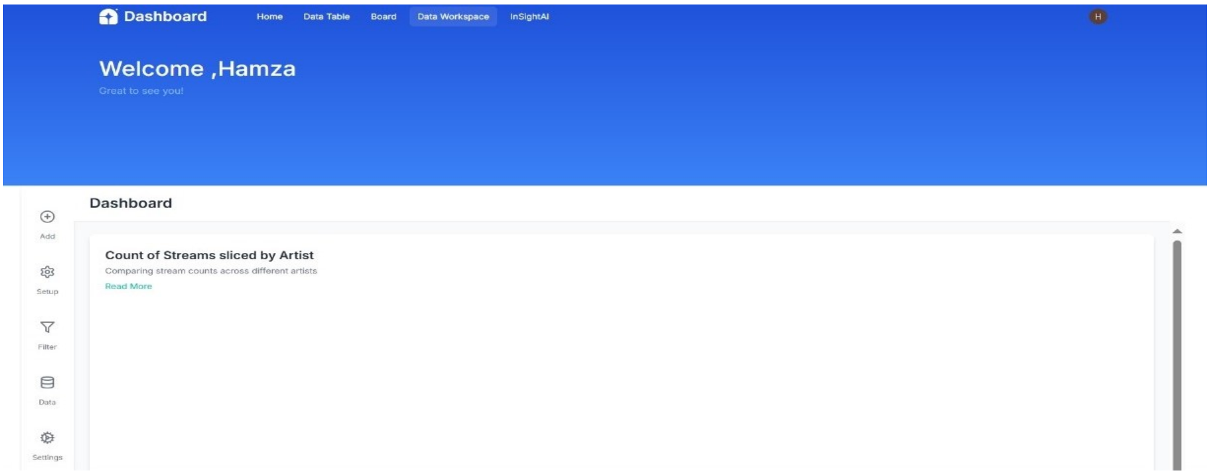

Figure 4.8: Dashboard Page: This figure represents the customizable dashboard for data visualization in InsightAI

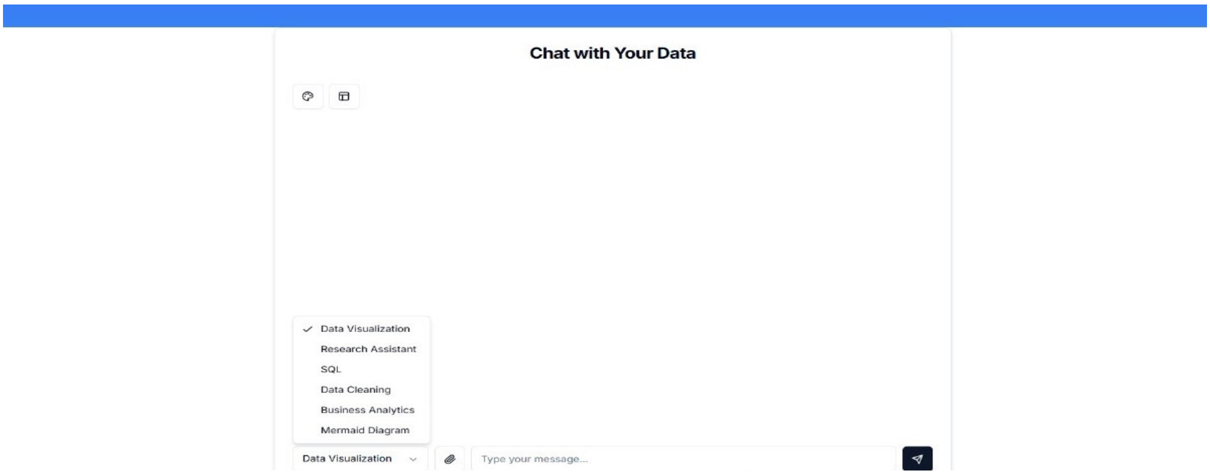

Figure 4.9: AI Chat Page: This figure represents the AI-powered chat interface for user interaction with InsightAI

4.9 Database Design

Our application presents users, datasets, analytics, visualizations, chats and reports in an efficient and comprehensible approach. All these entities can be easily interacted by the users, which helps the users analyze the data, generate input, and build simple and elegant reports. The pleasant thing of the system's user interface is that these interactions are meant to be as simple as possible and are intended to be as effective as possible. It makes users capable of decision making based on statistics and analytics.

4.9.1 ER Diagram

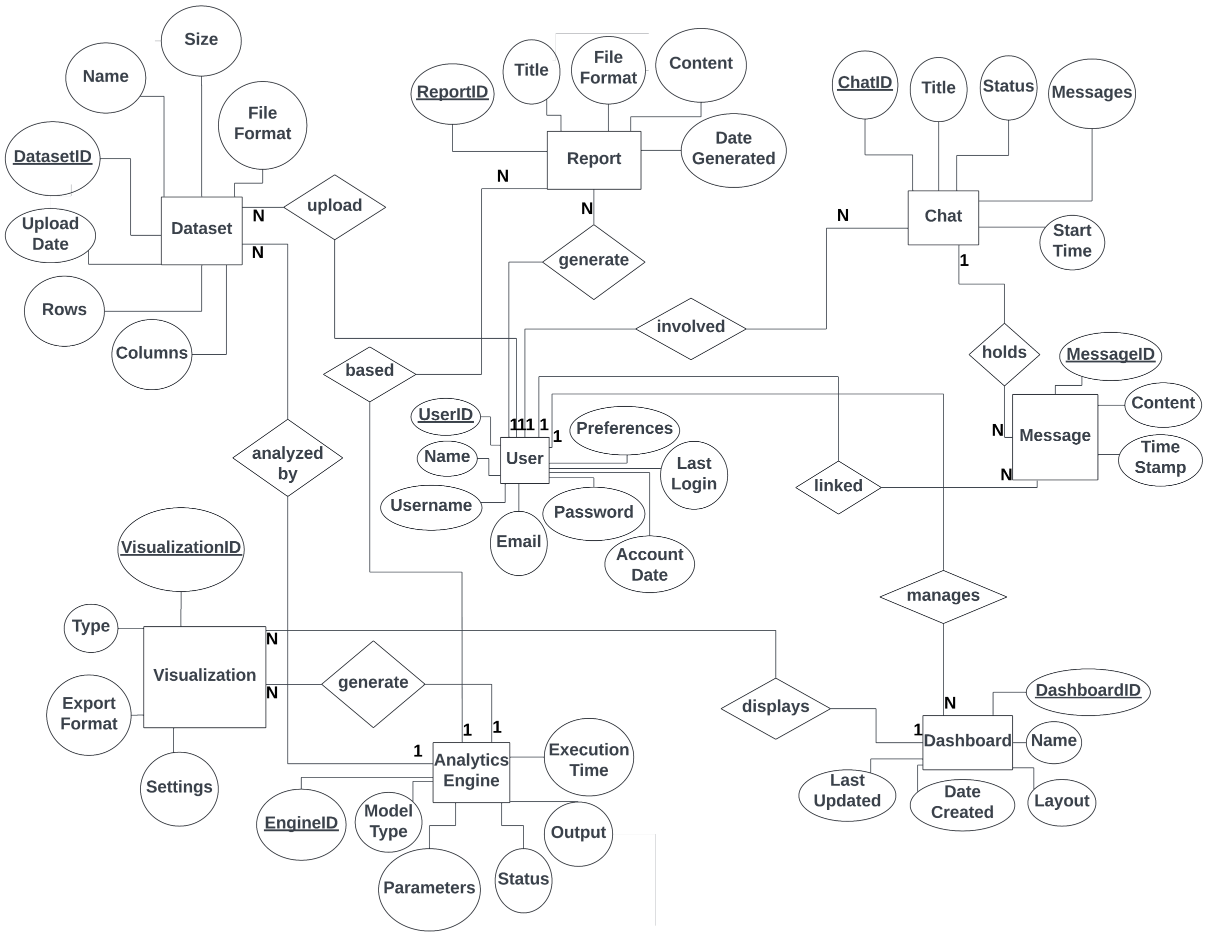

Figure 4.10: ER diagram: This ER diagram describes the database design of our application, showcasing the relationships between entities and their attributes

4.9.2 Data Dictionary

Table 4.1: Data Dictionary for User Entity: This table gives the details of the User entity its attributes, data type and relation

Attribute	Data Type	Relation	Nullable	Description
userId	int	-	No	Unique user identifier.
name	string	-	No	User's full name.
username	string	-	No	Unique username.
email	string	-	No	User's email address.
password	string	-	No	User's password.
accountDate	datetime	-	No	Account creation date.
preferences	JSON	-	Yes	Preferences in JSON format.
lastLogin	datetime	-	Yes	Last login timestamp.

Table 4.2: Data Dictionary for Dataset Entity: This table gives the details of the Dataset entity its attributes, data type and relation

Attribute	Data Type	Relation To	Nullable	Description
datasetId	int	User	No	Unique identifier for each dataset.
name	string	-	No	Name of the dataset.
size	float	-	No	Size of the dataset in megabytes.
fileFormat	string	-	No	Format of the dataset file (e.g., CSV, JSON).
uploadDate	datetime	-	No	Date when the dataset was uploaded.
rows	int	-	No	Number of rows in the dataset.
columns	int	-	No	Number of columns in the dataset.

Table 4.3: Data Dictionary for InsightAI Entity: This table gives the details of the InsightAI Engine entity its attributes, data type and relation

Attribute	Data Type	Relation To	Nullable	Description
engineId	int	Dataset	No	Unique identifier for each InsightAI engine.
modelType	string	-	No	Type of model used for analysis (e.g., regression, classification).
parameters	JSON	-	Yes	Configuration parameters for the analytics model.
status	string	-	No	Current status of the InsightAI engine (e.g., running, completed).
output	JSON	-	Yes	Output generated by the analytics process.
executionTime	float	-	No	Time taken to execute the analysis in seconds.

Table 4.4: Data Dictionary for Chat Entity: This table gives the details of the Chat entity its attributes, data type and relation

Attribute	Data Type	Relation To	Nullable	Description
chatId	int	User	No	Unique identifier for each chat session.
title	string	-	No	Title of the chat.
status	string	-	No	Current status of the chat (e.g., active, closed).
messages	string	-	Yes	List of messages exchanged in the chat.
startTime	datetime	-	No	Timestamp when the chat started.

Table 4.5: Data Dictionary for Message Entity: This table gives the details of the Message entity its attributes, data type and relation

Attribute	Data Type	Relation To	Nullable	Description
messageId	int	Chat	No	Unique identifier for each message.
content	string	-	No	Text content of the message.
timestamp	datetime	-	No	Timestamp when the message was sent.

Table 4.6: Data Dictionary for Visualization Entity: This table gives details of the Visualization entity its attributes, data type and relation

Attribute	Data Type	Relation To	Nullable	Description
visualizationId	int	InsightAI Engine	No	Unique identifier for each visualization.
type	string	-	No	Type of visualization (e.g., chart, graph).
exportFormat	string	-	No	Format for exporting the visualization (e.g., PNG, PDF).
settings	JSON	-	Yes	Configuration settings for the visualization.

Table 4.7: Data Dictionary for Report Entity: This table gives the details of the Report entity its attributes, data type and relation

Attribute	Data Type	Relation To	Nullable	Description
reportId	int	User	No	Unique identifier for each report.
title	string	-	No	Title of the report.
fileFormat	string	-	No	Format of the report file (e.g., PDF, DOCX).
content	JSON	-	Yes	Content of the report stored in JSON format.
dateGenerated	datetime	-	No	Date when the report was generated.

Table 4.8: Data Dictionary for Dashboard Entity: This table gives the details of the Dashboard entity its attributes, data type and relation

Attribute	Data Type	Relation To	Nullable	Description
dashboardId	int	User	No	Unique identifier for each dashboard.
name	string	-	No	Name of the dashboard.
layout	JSON	-	Yes	Layout configuration for the dashboard.
dateCreated	datetime	-	No	Date when the dashboard was created.
lastUpdated	datetime	-	No	Date when the dashboard was last updated.

4.10 Risk Analysis

Risk analysis is a vital part of any project's planning, identification of potential challenges and uncertainties that may affect the project's progress. By conducting a risk analysis on various factors, we proactively mitigate potential issues, ensuring a smooth development process and timely project delivery.

4.10.1 Hardware Failures

Although InsightAI dashboard is programmed to run on minimal hardware requirements, unexpected hardware failures or malfunctions can disrupt service availability. To reduce such a risk, regular hardware maintenance and backup systems are to be implemented to ensure a rapid recovery in the event of hardware failures.

4.10.2 Compatibility, Security and User Experience

The application should be able to work in a variety of web browsers although differences in browsers' rendering might cause incompatibilities. InsightAI dashboard uses industry best practices to protect user information and prevent data breaches. Our application depends on some platforms to provide features so its functionality may be impacted by outages or other interruptions in the third-party services. Intuitive design and an easy-to-use interface can help improve user experience. Our application aims to provide users with the highest quality of usability, desirability and usefulness in the data analysis sector.

4.11 Conclusion

The application outlined the comprehensive software requirements for our application covering both functional and non-functional aspects. It provided a detailed description of the features, use cases, quality attributes and hardware/software prerequisites essential for the successful development and deployment of the application. By clearly mentioning user interactions, data processes and system behaviors this chapter makes sure that all necessary steps are in place to deliver a robust, scalable and user-friendly data analysis platform. Furthermore, risks were identified and addressed to mitigate potential challenges. This can help in guiding the development process and creating a user-friendly platform with greatest possible usability

Chapter 5 High-Level and Low-Level Design

This chapter presents the high-level and low-level design of our application. The chapter mainly focuses on its core architectural components, design strategies and implementation details. The application is built to offer a comprehensive data analysis platform with features like advanced analytics, customizable visualizations and user-friendly interfaces. The chapter covers the applications modular architecture and design principles. Also, the considerations made to ensure scalability, performance and security of the application. The architecture and design concepts laid out in this chapter make sure that the application can handle complex data processing tasks efficiently while maintaining a focus on usability and responsiveness.

5.1 System Overview

InsightAI dashboard is designed as a robust and a user-friendly web-based application to provide users with seamless data analysis tools, business insights, and customizable visualizations. Its functionality ranges from basic data preprocessing, querying and visualization to advanced AI-based analytics and reporting. The application is responsible for processing datasets, generating reports, and rendering visualizations but still keeping an intuitive interface for interaction and customization. Key components include user management, data handling, InsightAI engine and visualization tools.

5.1.1 User Management

The user management component is responsible for handling all aspects related to user authentication, user registration and user sign in functionalities. By doing so, the application can make sure of that the users can securely log in, access personalized dashboards and maintain their own preferences within the system.

5.1.2 Data Handling

Data handling is a vital component that manages the input, storage and preprocessing of various file formats, including CSV, PDF, DOC, and more. This subsystem allows users to upload their files which are then cleaned and transformed into a format ready for analysis and visualization. Querying and filtering of data is also supported by the application so that users can focus on relevant information. The robust data handling mechanism ensures compatibility with different data sources and manage large amounts of data efficiently.

5.1.3 InsightAI Engine

The InsightAI engine is the core of our applications data processing capabilities. It is powered by machine learning and advanced statistical algorithms. The application analyzes datasets to provide meaningful insights, generate correlation matrices, and other statistical measures. The engine allows users to perform in-depth business analysis, generate reports and extract hidden patterns in data. By utilizing AI, the engine simplifies complex analysis processes giving the users actionable insights quickly and efficiently. It also allows users to generate automated reports that summarize key findings and recommendations from the data.

5.1.4 Visualization Tools

This subsystem is responsible for providing and displaying data in various forms such as charts, graphs, flowcharts and more. It allows users to interact with their data, create custom visualizations, and export them in different formats for reporting purposes. The application provides suers with multiple customization options enabling the user to adjust the visualization to his taste and meet specific business or presentation needs.

5.2 Design Considerations

For our application to be successful, we need to cater to every problem and dependency. So to make sure that our end user does not face any sort of problem while using our application. This section will describe many of the issues that need to be addressed before attempting to devise a complete design solution.

5.2.1 Assumptions and Dependencies

The assumptions and dependencies related with our web application that must be handled are listed below.

- To access the web application, users are required to have the updated version of their browsers installed on their machines.
- The user will have a consistent internet connection.
- The application is designed to operate seamlessly on major operating systems.
- We assume that users have basic computer literacy.
- The design considers the possibility of future updates and expansions to accommodate evolving

users who need to perform data analysis and visualization.

- User has knowledge of Basic English for using this web application.

5.2.2 General Constraints

Our application aims to provide comfort to users so that they can easily carry out their data analysis and visualization tasks without the need for hiring any data scientists or experts of the field. However, these are some constraints that can impact the performance of our application. These constraints are listed below:

5.2.2.1 Hardware and Software Environment

- No internet connection available.
- Users may experience difficulty accessing the application due to high internet traffic.
- Certain browser versions may not be supported, causing access issues.
- Application could be vulnerable to hacking and, in extreme cases, may be shut down due to a breach.

5.2.2.2 End-User Environment

- Some users may lack knowledge or experience in using the internet.
- Users may be unfamiliar with how to navigate and operate a web application.

5.2.2.3 Availability or Volatility of Resources

- A weak internet connection will not allow proper access to the application.
- Defective devices could result in users being unable to interact with the application effectively.

5.2.2.4 Interoperability Requirements

- The application must maintain a stable connection between the user and the database.
- Compatibility with a variety of browsers and operating systems is required.
- The application should support integration with third-party applications and services via Application Programming Interfaces (APIs).

5.2.2.5 Security Requirements

- Usernames and passwords must be stored in the database in encrypted form to ensure security.
- A unique token should be generated for each user during a session to maintain secure interactions.

5.2.2.6 Interface/Protocol Environments

- The system must provide an intuitive interface for easy navigation.
- The interface should be user-friendly and allow for seamless data input and display.

5.2.2.7 Performance Requirements

- The application should be capable of handling a large volume of users simultaneously without compromising performance.
- The system must be available at least 95 percent of the time within a 24-hour period.

5.2.3 Goals and Guidelines

InsightAI dashboard is designed with several key goals, guidelines, principles, and priorities to ensure its effectiveness and user satisfaction. The user interface should remain simple and intuitive despite the application's complex functionalities, following the KISS principle ("Keep It Simple Stupid!"). The application must balance performance and resource utilization to process large datasets effectively while maintaining responsiveness for the user. The user interface will emphasize ease of use with clear navigation, data visualization and customizable themes. User preferences and customization options will be prioritized.

Scrum methodology will be used as our development method. Scrum is an easy-to-use adaptable framework for leading and finishing complicated projects. The work will be broken into fixed-length iterations called sprints in Scrum. There shall be frequent reviews and changes to make the application produce useful and accurate results. A daily sprint meeting is scheduled to discuss the work that has to be done today. Furthermore, a sprint review meeting is scheduled after every sprint completion to review the progress, and a sprint retrospective meeting is scheduled before the start of a new sprint for the identification of sprint goals and any other changes that are required.

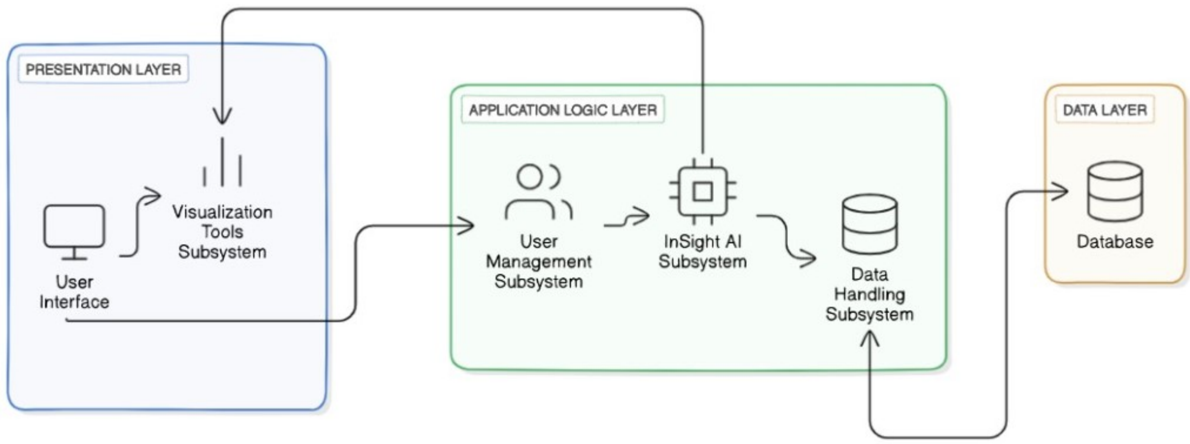

Figure 5.1: High-Level System Architecture Diagram: This picture contains the high-level system architecture of InsightAI dashboard that explains the deployment and data flow in application

5.3 System Architecture

5.3.1 Subsystem Architecture

The architecture is divided into four primary subsystems, each responsible for a specific aspect of the application:

5.3.1.1 User Management Subsystem

This subsystem handles the user registration, authentication, profile management and preference settings. It consists of the user database, authentication service and user profile manager. This subsystem communicates with the InsightAI Engine to manage user specific data access and preferences.

5.3.1.2 Data Handling Subsystem

This subsystem handles the data upload, storage, preprocessing and querying. Compatibility with various file formats is handled by this subsystem and it also handles data cleaning. It consists of file uploader, data transformer and query manager. This subsystem communicates with the InsightAI Engine for data analysis and with the Visualization Tools for presenting processed data.

5.3.1.3 InsightAI Engine Subsystem

This subsystem employs machine learning algorithms to analyze datasets, generate insights and create reports. It consists of machine learning models, statistical analysis tools and report generator. This subsystem receives data from the Data Handling Subsystem and sends insights to Visualization Tools for display.

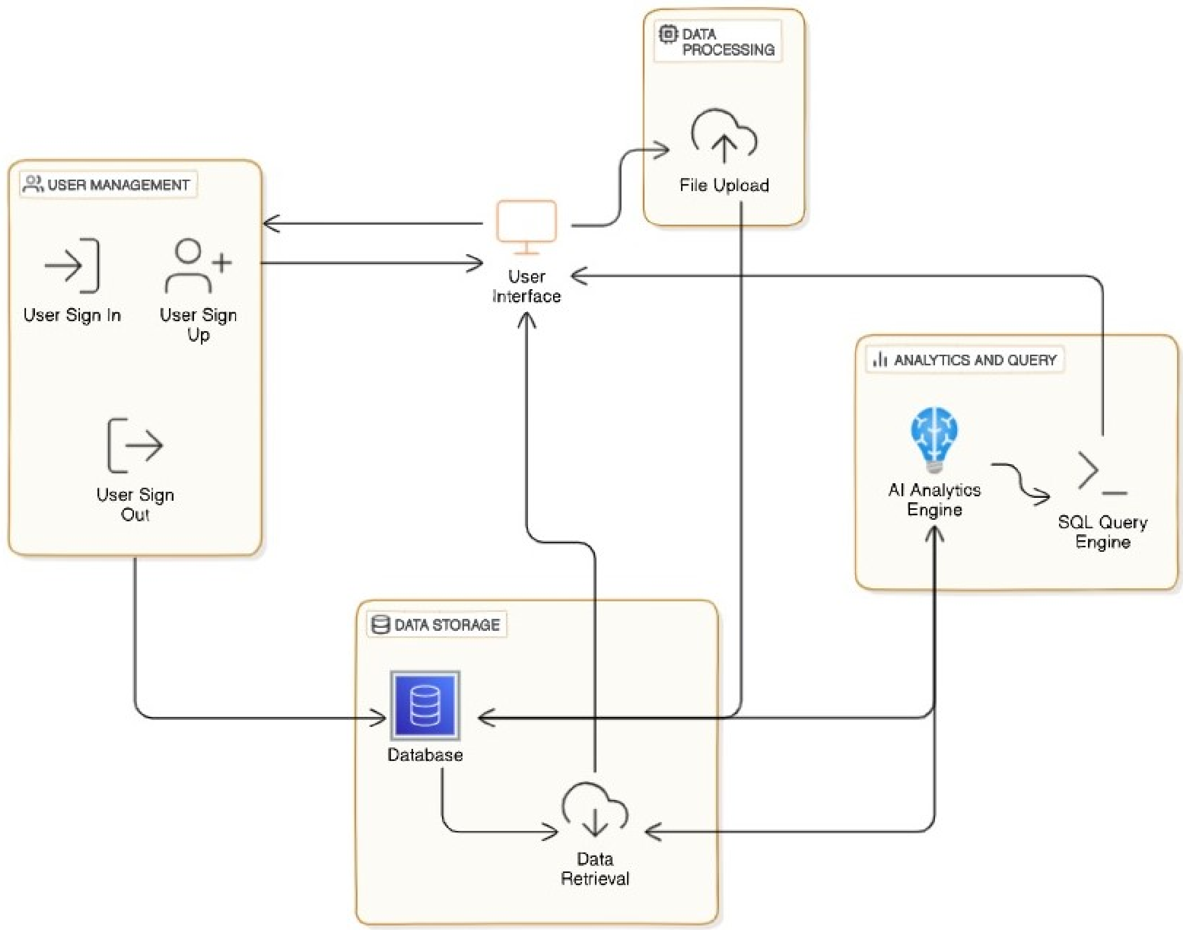

Figure 5.2: User Management Subsystem Diagram: This figure tells about the architecture of the user management subsystem, handling registration, authentication, and profile management for InsightAI users

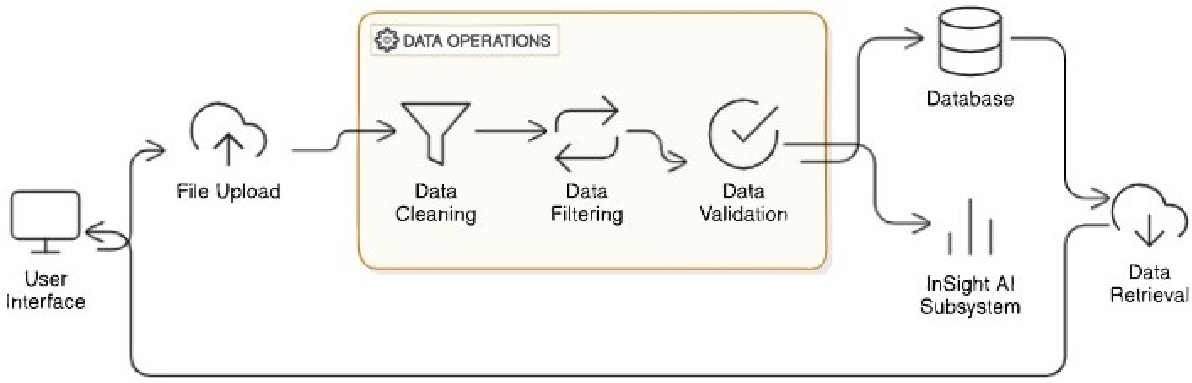

Figure 5.3: Data Handling Subsystem Diagram: This figure shows the data handling subsystem, responsible for data upload, preprocessing, and querying across multiple file formats in InsightAI

5.3.1.4 Visualization Tools Subsystem

This subsystem allows customizable visualization options and export features for data presentation. It consists of charting libraries, visualization editor and export manager. This subsystem works with

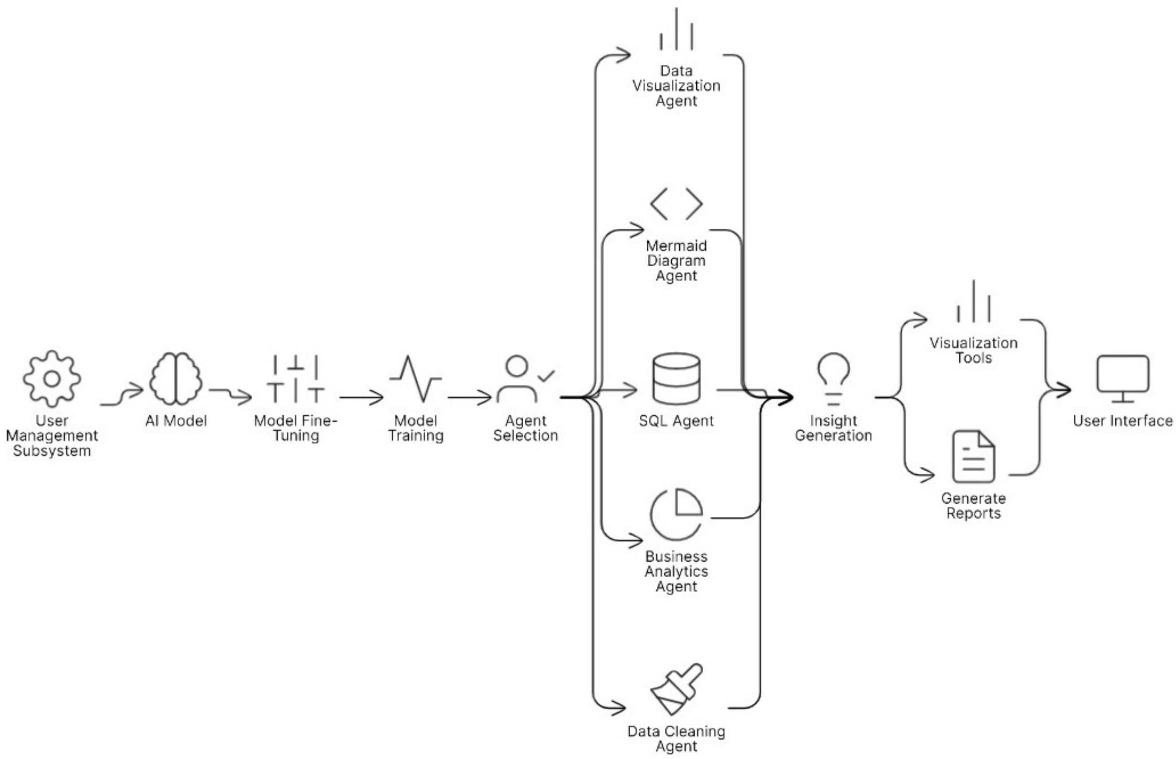

Figure 5.4: InsightAI Engine Subsystem Diagram: This figure represents the InsightAI Engine subsystem, showcasing how machine learning algorithms generate insights and reports from analyzed data

the InsightAI Engine to display insights and connects with the User Management subsystem to save customized visualizations.

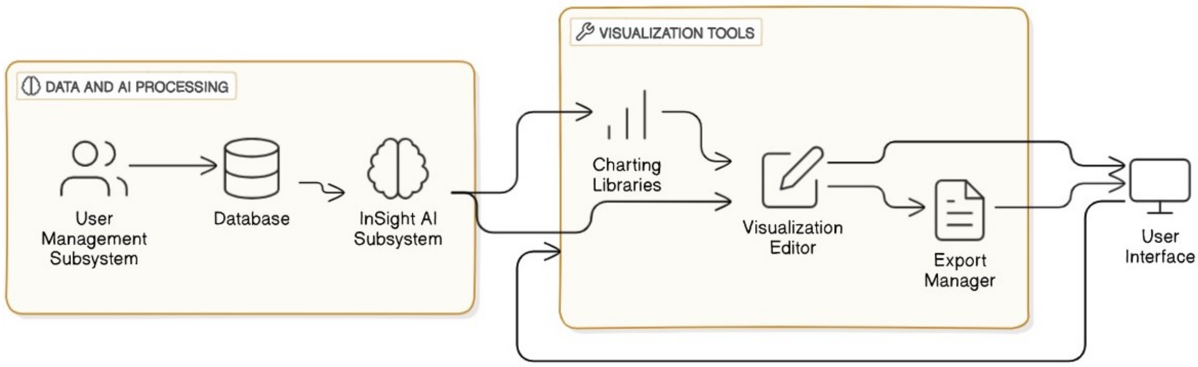

Figure 5.5: Visualization Tools Subsystem Diagram: This figure depicts the visualization tools subsystem, which facilitates customizable data presentations and exports insights generated by the InsightAI Engine

5.4 Architectural Strategies

The architectural strategies employed in our application aim to ensure efficient functionality, scalability and user satisfaction. Key strategies are outlined below:

5.4.1 Use of Microservices Architecture

Microservices architecture allows for development of independent services for user management, data handling, analytics and visualization. Each service can be independently deployed, scaled and also its flexibility and responsiveness can be enhanced to users needs. It allows for easier maintenance as updates can be made to individual services without affecting the entire application. New services can be added in the future without significant rework.

5.4.2 API-Driven Design

Application is designed to utilize RESTful APIs for communication between subsystems. Each subsystem uncovers specific APIs for interaction allowing other systems to integrate seamlessly. This approach ensures components can communicate effectively while allowing for future extensibility and integration with third-party services. This design choice was made to enhance interoperability and promote the reuse of existing components.

5.4.3 User-Centered Interface Design

User interface is designed keeping in mind usability and simplicity, adhering to the KISS principle. User feedback is used to make sure that the application user interface meets the demands of the users.

5.4.4 Scalability Considerations

Scalability was a important consideration in the architectural design. The application is designed to put up with increasing user loads and data volumes through load balancing and cloud-based resources.

5.4.5 Security Measures

A strong emphasis is placed on security and data privacy. Strategies such as encryption, secure API design, and compliance with data protection regulations are being implemented

5.5 Class Diagram

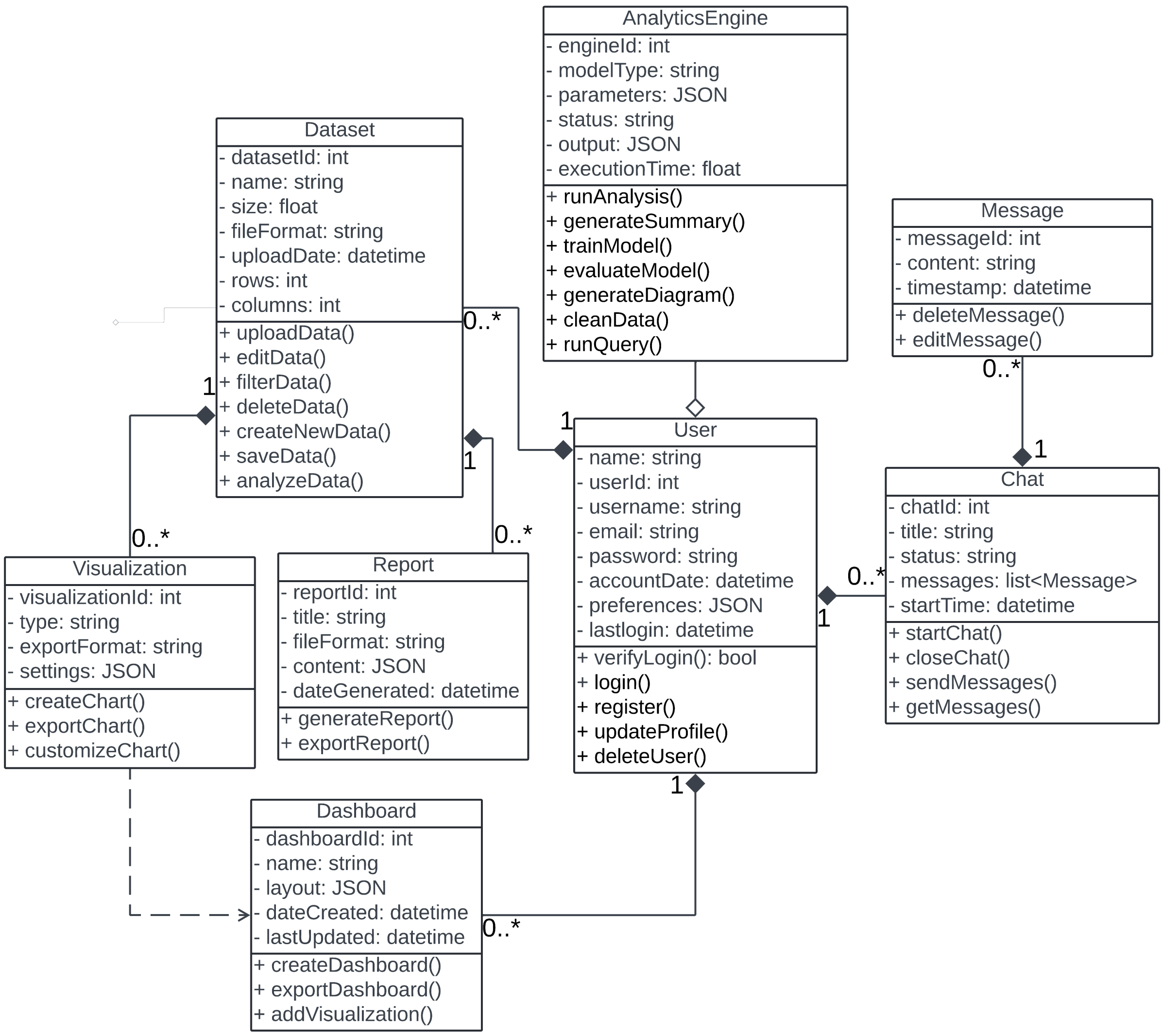

Figure 5.6: UML Class diagram: This figure represents UML Class Diagram for the functional and structural working of the InsightAI dashboard application

5.6 Policies and Tactics

This section outlines the key policies and tactics that will guide the design and implementation of our application. These decisions help with choosing tools, making trade-offs, following coding practices, and planning for testing and maintenance.

5.6.1 Codebase Organization

Policy/Tactic-1: Hierarchical Source Code Organization Our application's code is organized in a hierarchical structure, there is clear division of codes into logical components, files and directories.

Why? A well-structured codebase helps developers find what they need quickly and encourages modular design thus making collaboration smoother.

5.6.2 Development Tools

Policy/Tactic-2: Using Standard Tools Our application uses a set of standard development tools like compilers, interpreters, and libraries. These tools are choosen to make sure that everything works well together and performs reliably throughout the development.

Why? By using the same tools, developers can keep things constant this in turn reduces complexity of the problems and ensures a stable working environment.

5.6.3 Coding Guidelines

Policy/Tactic-3: Consistent Coding Practices Our application follows a set of coding guidelines which ensure that all the developers working on the project use the same style when writing code. This includes rules for naming things, indenting and adding comments.

Why? It is easier to read and understand consistent code. Collaboration is easier and the overall quality of the software is improved.

5.6.4 Testing Strategies

Policy/Tactic-4: Thorough Testing Plans Our application uses a detailed testing plan that includes unit, integration and system testing. Automated tools are also brought to work to make sure that everything works as expected.

Why? Testing helps catch bugs early and ensures the system works correctly. By the use of automated tests, the testing speed is increased and the test also provide quick feedback to developers.

5.6.5 Documentation Practices

Policy/Tactic-5: Clear Documentation Our application enforces on the importance of good documentation including inline comments in the code, API documentation and user guides. By doing so it ensures that all the users have access to the information they need.

Why? Good documentation makes it easier to understand, maintain and collaborate on the software. It's a key resource for both the development team and end-users.

5.6.6 Engineering Trade-offs

Policy/Tactic-6: Balancing Engineering Trade-offs When designing our application decisions often involve trade-offs between important factors such as system performance, resource usage and development time. These trade-offs are carefully analyzed to ensure that they align with the project's goals and requirements. Each and every decision is documented to provide clear reasoning behind the choices made.

Why? Balancing trade-offs is essential to make sure that the application meets performance standards, stays within resource limits and is developed efficiently. A thoughtful and careful evaluation of these factors helps in making informed and strategic design choices.

These policies and tactics come together to create a well-organized and reliable development environment for InsightAI dashboard. Especially focusing on clarity, consistency and quality of the application

5.7 Conclusion

This chapter outlines the high-level and low-level design for InsightAI telling about the system's architecture and key design decisions. Through user-friendly design and AI-powered data processing InsightAI offers a robust and customizable solution for data analysis and visualization to its users.

Chapter 6 Implementation and Test Cases

The implementation phase of our project is focused on bringing the theoretical concepts in to real functioning components. We used modern web development technologies and advanced machine learning techniques. The platform provides a seamless user interaction, efficient data handling and AI-driven insights. This chapter gives a detailed realization of the project's architecture with a main focus on the backend algorithms and subsystem interactions. The detailed analysis makes sure that the system is reliable and can handle growth while explaining the technologies used to achieve the project's goals.

6.1 Implementation

Our project is being developed using state-of-the-art web development technologies to ensure scalability, performance and user-friendliness. We have built the front end using Next.js, a React framework, while the back end is implemented with FlaskAPI or FastAPI to create a robust and efficient application programming interface (API). We have utilized SQLite as our primary relational database and have used VectorDB for enhanced data retrieval. For styling and interactivity we are using Tailwind CSS and TypeScript that will provide a visually appealing and responsive user interface.

6.1.1 Implementation of First Component: User Management Subsystem

The User Management Subsystem handles core functionalities like user registration, authentication and profile management. Clerk authentication middleware is being utilized in the backend for secure user sessions. The system employs hashing techniques for password security and session token management. User data is stored in SQLite which ensures lightweight and reliable database operations.

6.1.2 Implementation of Second Component: Data Handling Subsystem

The Data Handling Subsystem is responsible for managing the uploading, storage, preprocessing and querying of structured and unstructured data. A file uploader handles the uploading of files including multiple file formats such as CSV, Excel and JSON. Preprocessing includes data cleaning algorithms implemented using Pandas for missing value handling, duplicate removal and normalization. The raw data is transformed into queryable formats to maintain the compatibility with InsightAI Engine.

6.1.3 Implementation of Third Component: InsightAI Engine Subsystem

Machine learning models are used for predictive analytics and trend generation by the core analytical engine. The engine is trained to generate insights from the processed data. Statistical analysis modules also generate insights that are then compiled in the form of a report by the report generator.

6.1.4 Implementation of Fourth Component: Visualization Tools Subsystem

This subsystem generates interactive and customizable visualizations. It employs libraries like ReCharts, Plotly and more for creating charts, graphs, and flow diagrams. Users can modify visual properties by the help of the visual editor. Furthermore, users are able to save charts and visualizations in multiple formats.

6.2 Conclusion

In the implementation phase, the foundation of our web application InsightAI Dashboard was established. The high-level architecture diagrams were transformed into functioning components. Each and every subsystem was carefully planned and developed to ensure smooth integration and robust functionality. From user authentication to AI-driven insights and dynamic visualizations, the implementations align with the project's scope and goals.

Chapter 7 Conclusions

This chapter provides an overview of the work performed during FYP-1 including the objectives achieved, key findings and the results obtained. It also discusses the challenges that we faced during the project. Finally, it outlines a plan for the FYP-2.

7.1 Summary of Work Done

The InsightAI Dashboard project worked towards the creation of an AI-powered and user-friendly data visualization and analysis platform. Significant progress was achieved in defining the system architecture and in implementing the core functionalities of the web application. Key subsystems such as User Management, Data Handling, the InsightAI Engine and Visualization Tools were developed conceptually with a prototype demonstrating basic interactions. For scalability and efficiency, technologies such as Next.js, FlaskAPI, SQLite and VectorDB were used in the implementation of the web application. Initial algorithms for data preprocessing and visualization were also designed and tested.

7.2 Findings and Results

The prototype ensured the feasibility of the following:

- Data analysis and visualization powered by AI.
- Handling of structured and unstructured datasets.
- Providing intuitive user interfaces for nontechnical users.
- Support for automated data cleaning and large dataset uploads.

Even though the scope was partially covered, the foundation goals such as the system architecture and core functionalities were accomplished. The implementation of the key algorithms including data preprocessing and visualization rendering proved to be effective in addressing the project's objectives.

7.3 Challenges Faced

Several challenges emerged:

- Integrating diverse technologies into a cohesive system required significant effort.
- Optimizing machine learning algorithms to minimize errors while ensuring scalability posed difficulties.

- Constraints in time and resources limited the implementation of advanced AI features.
- AI responses are not always accurate and sometimes do provide incorrect outputs this impacts the reliability of the application.

7.4 Plan for FYP-2

- Automate the creation of detailed and customized reports for users based on the analyzed datasets.
- Develop a agent to identify and correct the inconsistencies, outliers and missing values in the data.
- Build an agent that will perform statistical analysis on the data to generate actionable insights.
- Enable users to design and personalize dashboards according to their own liking.
- Complete implementation of incomplete or features that are left.
- Extend the prototype into a fully functional system.
- Conduct rigorous testing, including performance optimization and error analysis.
- Finalize the documentation, showcasing the application's value and usability.

7.5 Final Conclusion

To conclude, FYP-1 has laid a strong foundation for our project InsightAI Dashboard by achieving significant progress in system architecture, feature implementation and prototype development. The project is well laid to achieve its full potential in FYP-2 by addressing all the challenges faced and expanding capabilities to meet all objectives. With further processing, InsightAI Dashboard has the potential to become a transformative tool in data visualization and analysis and disrupt the market.

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